Full-Jacobian IC on the transferred instance (O3)

Full-Jacobian index calculus on the transferred instance (open case O3): results

Status: `[pcc:unverified]`. This is one session on a shared 4-vCPU VM with no isolation. It has no signed pcc claim and is not a row in the IC leaderboard or in `benchmarks/README.md`. Protocol: `PROTOCOL.md`, frozen before any data (`frozen.sha256`; `sha256sum -c` reports OK). Raw output: `full_jacobian.json` and `run.log`. Full tables: `tables.md`. Figure: `full_jacobian.svg`.

Question

T5–T7 in `research/chained_transfer_proof_20261112/PROOF.md` show that an index calculus whose relations lie in the transferred subgroup `S` of the genus-2 Jacobian degenerates to a collision search on `E`. O3 is the case those theorems leave open: an IC that draws relations from the whole `J_C(F_p)`, Gaudry-style. The question is whether that IC, run on the transferred instance, gets anywhere near the matched negation-map rho on `E1`.

Arms

arm	what it is
A	whole-J IC, full relations: walk `aG' + bT' + kZ` with `Z` a random whole-J generator, one extra unknown `ζ = log_{G'}(hZ)`
B	whole-J IC, double large prime
A_S, B_S	the same IC restricted to `S` (the T5–T7 controls)
rho on E1 (negation)	the matched reference, with set-up separated
rho on J (negation)	control: rho in the Jacobian's `r`-part

All arms are certified per log, with 0 bad relations and 0 exhausted runs. Every arm's group operations, linear-algebra multiplications and additions are priced on the §5a band (0 / 0.1 / 1 GAE per operation) and at the calibrated reading (0 / w_cal / 10·w_cal). The transfer itself is not charged (trapdoor convention).

Result

Every cell, at every prime from 211 to 51,203 and for all four IC arms, is `above` the negation rho on E1, both at the band and at the calibrated reading. The distance remaining at `w = 0.1` (arm total / rho total):

p	r	A	B	A_S	B_S
211	109	7.0e3	406	114	51
419	113	2.4e4	1.8e3	246	133
821	71	5.6e4	4.3e3	434	240
1619	401	4.4e4	1.7e3	1.5e3	220
3203	313	3.4e4	1.2e3	3.9e3	556
6421	3191	2.4e4	626	1.1e4	649
12841	3169	1.2e5	881	2.7e4	920
25601 (stage 2)	127	6.2e6	5.7e3	2.7e4	4.9e3
51203 (stage 2)	1831	7.5e6	4.7e3	1.8e5	6.5e3

At `w = 0`, with group operations only, the smallest ratio is 18.7, for A at p = 211. Across the stage-1 primes, the cheapest whole-J arm, B, needs 18× to 54× the rho group operations before any linear algebra is charged.

Verdict under the frozen decision rule: unresolved. Rule F1, “every cell above”, holds for both arms. The second condition, a lower 95% bound above 0.2 on the slope of `ln ratio` on `ln p` at `w = 0.1`, fails:

arm	slope (stage 1, n = 7)	95% interval
A	0.421	[−0.070, 0.912]
B	−0.053	[−0.654, 0.549]

No cell is `below` at any price, so the “advantage seen” branch is not triggered either. The rule’s blocking-cell list is empty: the result is unresolved because of the slope, not because of any cell.

Why the slope criterion fails

The protocol assumed rho on E1 would cost about √p. It costs about √r, and the selected `r` does not grow with `p`: it is 71 at p = 821, 3191 at p = 6421 and 127 at p = 25601. The fitted rho exponent on `ln p` is 0.236, with 95% interval [0.120, 0.352]. Against `ln r` it is 0.234, flattened at these sizes by the set-up and the distinguished-point overhead. So the ratio’s denominator moves with `r` almost independently of `p`, and seven points cannot fix a slope in `p`. This is a defect in the instance selection, which did not bind `r` to `p`, and not a sign of an advantage. The secondary, unfrozen fit of `ln(group ops / √r)` on `ln p` gives 0.420 for A and 0.292 for B, both positive.

The B slope is also noisy for a second reason: B’s cost at `w = 0.1` is dominated by the linear-algebra additions, which change by large factors between primes as the number of large-prime merges changes.

Scorecard of the frozen predictions

id	prediction	outcome
F1	every A and B cell above at band and calibrated, 7 stage-1 primes	**held** (7/7 for each)
F2	ratio slope in [0.3, 0.7] for A and B; lower 95% end > 0.2	**failed**: A 0.421 is in range, B −0.053 is not; neither lower end clears 0.2

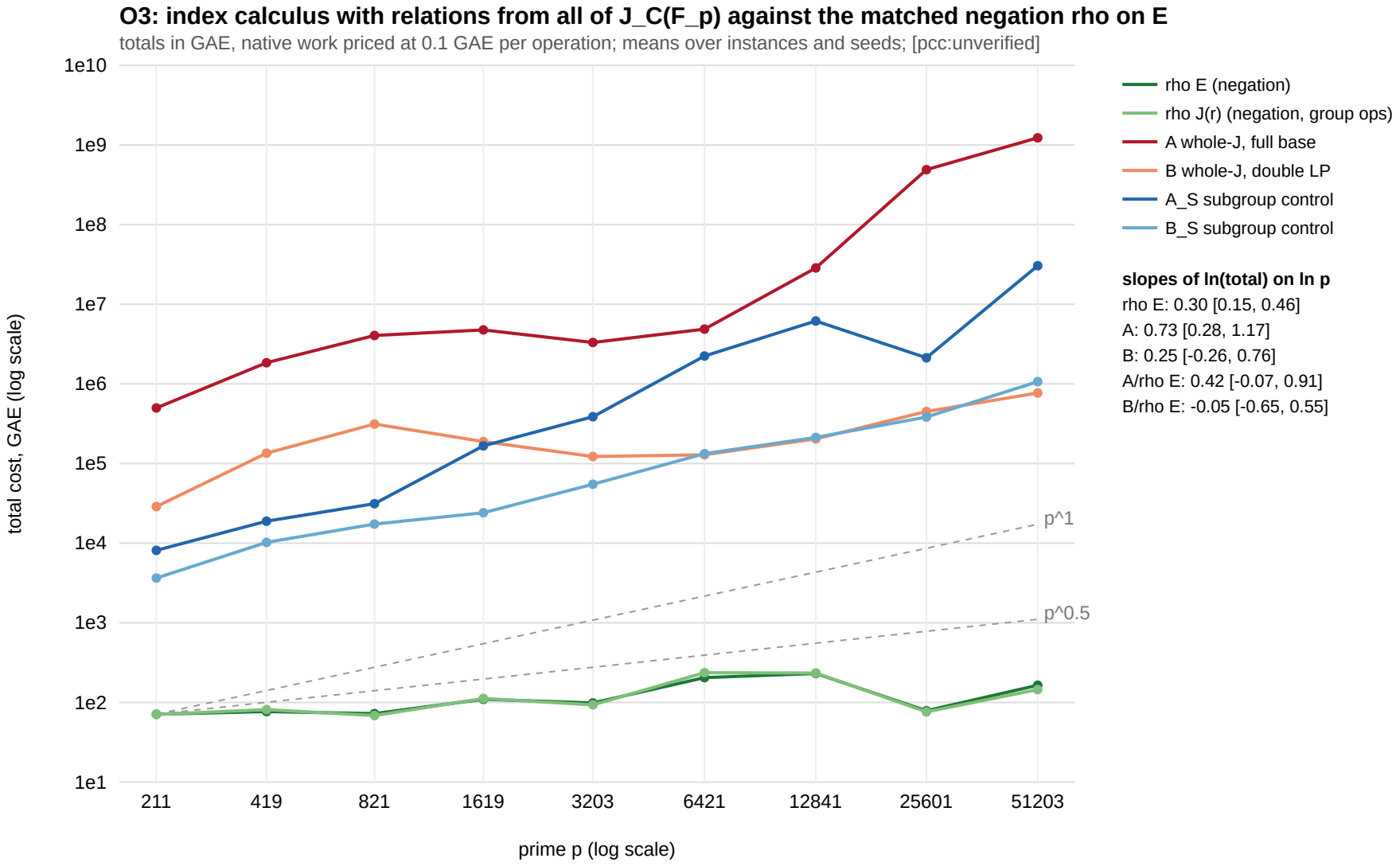
id	prediction	outcome
F3	group-op exponent A, B in [0.8, 1.25]; rho on E1 in [0.4, 0.6]	**failed**: A 0.552, B 0.444, rho 0.236 (r not bound to p)
F4	B cheaper than A at w = 1, p = 12841	**held** (1.96e6 vs 2.85e8)
F5	A within [0.4, 2.5]× A_S in group ops, p = 12841	**held** (0.71)
F6	deg-2 smooth fraction in [0.4, 0.6]	**held**: per-prime means 0.483–0.536 for A and B
F7	every log certified, no instance skipped at stage 1	**held** (12/12 per arm per prime)
F8	stage 2 completes p = 25601 and p = 51203	**held**, on the rerun: the first run was killed at p = 51203 before p = 102407; the rerun used `--stage2 25601,51203` under the same open record

Reading

No evidence of an advantage: the whole-J arms A and B are above the matched rho by factors of 406 to 7.5e6 at `w = 0.1`, at every prime measured. That agrees with the best-known-algorithm comparison in PROOF.md O3 ($\tilde{O}(p^{2-2/g})$ against $\sqrt{r}$). But the frozen rule needed to call O3 “closed by data” is not met, because of the instance-selection defect above, so O3 remains open in PROOF.md. A rerun that would decide it fixes $r \approx p$, so that `#E1 / r` stays small, and freezes that choice in a new protocol. The current selection does not bind `r` to `p` (h = #J / r ranges from 452 to 5.1e6).

Limits: toy size (p ≤ 51,203); one shared VM with no isolation (wall times are indicative only); stage 2 has 2 IC seeds and 1 instance per prime.

Figure



Group operations and totals per prime (means; native priced at w = 0.1 GAE)

p	r	#J	rho E (neg) ops	rho E total w=0.1	A ops	A total w=0.1	B ops	B total w=0.1	A_S ops	B_S ops	rho J(r) ops
211	109	49268	64.0	71.1	1194	498213	1150	28889	764	698	63.7
419	113	187128	68.5	76.9	1715	1.85e+06	1676	134991	1589	1781	72.1
821	71	657744	65.7	72.4	2972	4.06e+06	2927	313496	2264	2338	63.0
1619	401	2502240	89.8	109.5	3809	4.77e+06	4188	188843	3802	2673	89.9
3203	313	9834460	83.6	98.7	3944	3.31e+06	3133	122458	7334	5842	79.9
6421	3191	40347004	145.3	205.5	7289	4.87e+06	5508	128723	10691	9452	158.0

p	r	#J	rho E (neg) ops	rho E total w=0.1	A ops	A total w=0.1	B ops	B total w=0.1	A_S ops	B_S ops	rho J(r) ops
12841	3169	163824624	161.3	231.6	14048	2.85e+07	8675	204121	19884	13526	160.5
25601	127	649528800	70.1	78.7	30132	4.89e+08	17562	451664	56189	32552	68.2
51203	1831	2619589728	125.9	164.8	58423	1.23e+09	30903	769174	101329	71922	115.8

Ratio to the matched negation rho on E (band w = 0, 0.1, 1) and calibrated reading (0, w_cal, 10 w_cal)

p	arm	ratio w=0	ratio w=0.1	ratio w=1	cal 0	cal w_cal	cal 10 w_cal	class (band)	class (cal)
211	A	18.7	7002.7	36751.3	18.7	603.5	5159.0	above	above
211	B	18.0	406.1	2059.2	18.0	50.3	302.6	above	above
211	A_S	11.9	114.0	548.8	11.9	20.4	86.4	above	above
211	B_S	10.9	51.4	223.6	10.9	14.2	40.0	above	above
419	A	25.0	24034.8	121406.1	25.0	2090.1	18058.9	above	above
419	B	24.5	1756.0	8778.1	24.5	173.2	1323.7	above	above
419	A_S	23.2	245.6	1147.3	23.2	42.2	188.9	above	above
419	B_S	26.0	133.2	568.1	26.0	35.0	105.0	above	above
821	A	45.2	56094.7	307163.3	45.2	4694.3	41529.2	above	above
821	B	44.5	4332.3	23538.9	44.5	400.0	3215.9	above	above
821	A_S	34.4	433.6	2221.4	34.4	67.3	328.1	above	above
821	B_S	35.6	240.4	1157.9	35.6	52.4	185.3	above	above
1619	A	42.4	43554.8	165860.4	42.4	4014.4	33337.5	above	above
1619	B	46.7	1724.0	6438.7	46.7	199.5	1328.3	above	above
1619	A_S	42.4	1520.8	5676.3	42.4	177.1	1171.9	above	above
1619	B_S	29.8	219.9	754.2	29.8	47.0	174.0	above	above
3203	A	47.2	33510.4	140715.6	47.2	3027.8	25742.0	above	above
3203	B	37.5	1240.9	5096.4	37.5	144.5	960.3	above	above
3203	A_S	87.8	3924.5	16216.1	87.8	429.2	3030.9	above	above
3203	B_S	69.9	556.3	2114.6	69.9	112.9	440.9	above	above
6421	A	50.2	23720.3	65123.6	50.2	2418.6	18460.5	above	above
6421	B	37.9	626.4	1655.8	37.9	96.7	494.9	above	above
6421	A_S	73.6	10884.1	29793.5	73.6	1155.1	8480.8	above	above
6421	B_S	65.1	648.9	1670.2	65.1	123.3	517.9	above	above
12841	A	87.1	123232.2	329865.1	87.1	12562.1	97355.7	above	above
12841	B	53.8	881.2	2269.5	53.8	137.6	707.1	above	above
12841	A_S	123.3	26654.6	71173.1	123.3	2810.9	21079.1	above	above
12841	B_S	83.9	919.8	2322.5	83.9	168.5	743.8	above	above
25601	A	429.7	6212348.5	31414419.9	429.7	490581.5	4423524.7	above	above
25601	B	250.4	5741.3	28018.1	250.4	682.8	4152.2	above	above
25601	A_S	801.3	27077.1	133679.7	801.3	2871.8	19485.4	above	above
25601	B_S	464.2	4868.1	22735.0	464.2	810.1	3585.4	above	above
51203	A	463.9	7475818.7	23941937.9	463.9	674451.5	5503674.3	above	above
51203	B	245.4	4667.3	14407.6	245.4	643.2	3493.3	above	above
51203	A_S	804.6	184681.9	589712.2	804.6	17380.3	136147.5	above	above
51203	B_S	571.1	6471.9	19469.6	571.1	1101.0	4897.8	above	above

Arm detail (means over runs)

p	arm	runs	certified	exhausted	columns	trials	relations	core cols	attempts	rel mul	la mul	smooth rate	wall s
211	A	12	12	0	115	552	289	51	11.75	77428	4.89e+06	0.536	0.07
211	B	12	12	0	30	519	271	13	11.67	69963	207431	0.540	0.00
211	A_S	12	12	0	115	232	119	34	1.17	34066	39412	-	0.00
211	B_S	12	12	0	30	203	106	4	1.00	28077	1475	-	0.00
419	A	12	12	0	222	1056	521	90	19.58	160167	1.83e+07	0.510	0.25
419	B	12	12	0	41	976	483	15	20.33	142554	1.19e+06	0.513	0.01
419	A_S	12	12	0	222	604	228	38	1.33	81674	91199	-	0.00
419	B_S	12	12	0	41	692	242	2	1.00	83363	1241	-	0.00
821	A	12	12	0	400	2261	1105	146	36.17	391324	4.02e+07	0.490	0.57
821	B	12	12	0	58	2102	1045	13	49.17	356666	2.75e+06	0.491	0.03
821	A_S	12	12	0	400	908	406	34	1.25	158452	132650	-	0.00
821	B_S	12	12	0	58	940	412	1	1.08	149490	1084	-	0.00
1619	A	12	12	0	771	2944	1432	217	18.33	558261	4.71e+07	0.485	0.67
1619	B	12	12	0	81	2948	1447	13	61.50	533242	1.31e+06	0.486	0.02
1619	A_S	12	12	0	771	1808	810	106	2.00	339448	1.29e+06	-	0.03
1619	B_S	12	12	0	81	1214	557	1	1.00	212730	1392	-	0.00
3203	A	12	12	0	1533	3308	1598	744	1.83	665140	3.24e+07	0.483	0.47
3203	B	12	12	0	114	2440	1179	36	7.42	449093	744151	0.483	0.01
3203	A_S	12	12	0	1533	3525	1610	130	2.00	683163	3.12e+06	-	0.05
3203	B_S	12	12	0	114	2771	1249	1	1.00	488660	1895	-	0.01
6421	A	12	12	0	3140	6401	3142	1374	1.00	1.47e+06	4.72e+07	0.491	0.68
6421	B	12	12	0	161	4569	2246	51	6.67	955330	276819	0.492	0.01
6421	A_S	12	12	0	3140	6824	3311	413	2.08	1.54e+06	2.07e+07	-	0.31
6421	B_S	12	12	0	161	5986	2898	1	1.00	1.24e+06	2617	-	0.01
12841	A	12	12	0	6378	13025	6460	2958	1.25	3.2e+06	2.82e+08	0.496	4.06
12841	B	12	12	0	227	7718	3821	31	3.58	1.73e+06	221852	0.495	0.02
12841	A_S	12	12	0	6378	12964	6697	598	2.00	3.25e+06	5.83e+07	-	0.88
12841	B_S	12	12	0	227	8708	4494	1	1.00	1.99e+06	3626	-	0.02
25601	A	2	2	0	12683	29199	14587	7134	4.00	7.29e+06	4.88e+09	0.500	71.32
25601	B	2	2	0	321	16766	8394	99	1.00	3.89e+06	452408	0.501	0.04
25601	A_S	4	4	0	12683	25838	13318	73	2.00	6.55e+06	1.42e+07	-	0.28
25601	B_S	4	4	0	321	14936	7760	1	1.00	3.5e+06	4951	-	0.04
51203	A	2	2	0	25579	56690	28139	12839	3.00	1.47e+07	1.23e+10	0.496	180.73
51203	B	2	2	0	453	29674	14746	46	1.00	7.09e+06	294162	0.497	0.07
51203	A_S	4	4	0	25579	62973	26859	736	2.00	1.5e+07	2.88e+08	-	4.48
51203	B_S	4	4	0	453	44626	19642	1	1.00	9.94e+06	7047	-	0.11

Selection, order data and calibration

p	candidate	skipped (by reason)	n1 = #E1	n2 = #E2	h = #J/r	r^2 divides #J	ord(Z)/#J (3 inst.)	t_add E (ns)	t_add J (ns)	w_cal mul_p (J)	w_cal mul_r (J)
211	9	{“bielliptic_rejected”: 6, “r_below_50”: 3}	218	226	452	False	0.50, 0.25, 0.25	103	588	0.0077	0.0077

p	candidate	skipped (by reason)	n1 = #E1	n2 = #E2	h = #J/r	r^2 divides #J	ord(Z)/#J (3 inst.)	t_add E (ns)	t_add J (ns)	w_cal mul_p (J)	w_cal mul_r (J)
419	27	{“bielliptic_rejected”: 18, “r_below_50”: 9}	452	414	1656	False	0.33, 0.06, 0.50	107	592	0.0078	0.0078
821	2	{“bielliptic_rejected”: 1, “r_divides_e2_order”: 1}	852	772	9264	False	0.25, 0.12, 0.12	117	604	0.0076	0.0076
1619	1	{“bielliptic_rejected”: 1}	1604	1560	6240	False	0.17, 0.03, 0.17	122	604	0.0076	0.0076
3203	12	{“bielliptic_rejected”: 10, “r_below_50”: 2}	3130	3142	31420	False	0.50, 0.50, 0.25	124	601	0.0077	0.0077
6421	2	{“bielliptic_rejected”: 1, “r_below_50”: 1}	6382	6322	12644	False	0.50, 0.50, 0.25	134	632	0.0073	0.0073
12841	2	{“bielliptic_rejected”: 1, “r_below_50”: 1}	12676	12924	51696	False	0.25, 0.08, 0.25	141	634	0.0073	0.0073
25601	1	{“r_divides_e2_order”: 1}	25400	25572	5114400	False	0.25	152	648	0.0071	0.0071
51203	8	{“bielliptic_rejected”: 8}	51268	51096	1430688	False	0.06	144	653	0.0071	0.0071

Fitted exponents, stage 1 primes (OLS slope of ln y on ln p, 95% t interval)

quantity	slope	95% interval	n
A_S_group_ops	0.771	[0.709, 0.834]	7
A_S_ratio_to_rho_E_w0	0.535	[0.383, 0.688]	7
A_S_ratio_to_rho_E_w0.1	1.366	[1.241, 1.491]	7
A_S_ratio_to_rho_E_w1	1.206	[1.122, 1.291]	7
A_S_total_w0	0.771	[0.709, 0.834]	7
A_S_total_w0.1	1.670	[1.417, 1.924]	7
A_S_total_w1	1.694	[1.440, 1.949]	7
A_group_ops	0.552	[0.423, 0.682]	7
A_ratio_to_rho_E_w0	0.316	[0.177, 0.456]	7
A_ratio_to_rho_E_w0.1	0.421	[-0.070, 0.912]	7
A_ratio_to_rho_E_w1	0.238	[-0.314, 0.789]	7
A_total_w0	0.552	[0.423, 0.682]	7
A_total_w0.1	0.725	[0.282, 1.169]	7
A_total_w1	0.726	[0.282, 1.169]	7
B_S_group_ops	0.687	[0.548, 0.826]	7
B_S_ratio_to_rho_E_w0	0.450	[0.250, 0.651]	7
B_S_ratio_to_rho_E_w0.1	0.661	[0.465, 0.857]	7
B_S_ratio_to_rho_E_w1	0.510	[0.228, 0.792]	7
B_S_total_w0	0.687	[0.548, 0.826]	7
B_S_total_w0.1	0.965	[0.837, 1.094]	7
B_S_total_w1	0.998	[0.871, 1.126]	7
B_group_ops	0.444	[0.286, 0.603]	7
B_ratio_to_rho_E_w0	0.208	[0.023, 0.394]	7
B_ratio_to_rho_E_w0.1	-0.053	[-0.654, 0.549]	7
B_ratio_to_rho_E_w1	-0.240	[-0.924, 0.444]	7

quantity	slope	95% interval	n
B_total_w0	0.444	[0.286, 0.603]	7
B_total_w0.1	0.252	[-0.256, 0.759]	7
B_total_w1	0.248	[-0.268, 0.763]	7
rho_E_negation_group_ops	0.236	[0.120, 0.352]	7
rho_E_negation_total_w0.1	0.304	[0.150, 0.459]	7
rho_J_negation_group_ops	0.240	[0.095, 0.385]	7

Fitted exponents, stage 1 and 2 primes (OLS slope of ln y on ln p, 95% t interval)

quantity	slope	95% interval	n
A_S_group_ops	0.865	[0.780, 0.951]	9
A_S_ratio_to_rho_E_w0	0.743	[0.538, 0.948]	9
A_S_ratio_to_rho_E_w0.1	1.308	[1.159, 1.458]	9
A_S_ratio_to_rho_E_w1	1.233	[1.144, 1.322]	9
A_S_total_w0	0.865	[0.780, 0.951]	9
A_S_total_w0.1	1.463	[1.159, 1.767]	9
A_S_total_w1	1.479	[1.169, 1.788]	9
A_group_ops	0.678	[0.553, 0.804]	9
A_ratio_to_rho_E_w0	0.556	[0.337, 0.775]	9
A_ratio_to_rho_E_w0.1	1.108	[0.470, 1.745]	9
A_ratio_to_rho_E_w1	1.017	[0.287, 1.748]	9
A_total_w0	0.678	[0.553, 0.804]	9
A_total_w0.1	1.262	[0.758, 1.767]	9
A_total_w1	1.263	[0.758, 1.768]	9
B_S_group_ops	0.778	[0.670, 0.887]	9
B_S_ratio_to_rho_E_w0	0.656	[0.450, 0.861]	9
B_S_ratio_to_rho_E_w0.1	0.824	[0.647, 1.000]	9
B_S_ratio_to_rho_E_w1	0.756	[0.485, 1.028]	9
B_S_total_w0	0.778	[0.670, 0.887]	9
B_S_total_w0.1	0.979	[0.899, 1.058]	9
B_S_total_w1	1.002	[0.923, 1.081]	9
B_group_ops	0.551	[0.428, 0.673]	9
B_ratio_to_rho_E_w0	0.428	[0.214, 0.642]	9
B_ratio_to_rho_E_w0.1	0.222	[-0.175, 0.618]	9
B_ratio_to_rho_E_w1	0.127	[-0.360, 0.615]	9
B_total_w0	0.551	[0.428, 0.673]	9
B_total_w0.1	0.377	[0.088, 0.666]	9
B_total_w1	0.373	[0.080, 0.666]	9
rho_E_negation_group_ops	0.123	[-0.009, 0.254]	9
rho_E_negation_total_w0.1	0.155	[-0.017, 0.327]	9
rho_J_negation_group_ops	0.113	[-0.032, 0.258]	9

Secondary, not frozen: dependence on r (rho on E costs about sqrt(r), r varies from prime to prime)

- slope of ln(rho E negation ops) on ln r: 0.234

- A: group ops / sqrt(r) per prime: 211: 114.4, 419: 161.4, 821: 352.7, 1619: 190.2, 3203: 222.9, 6421: 129.0, 12841: 249.5, 25601: 2673.8, 51203: 1365.3
- A: slope of ln(group ops / sqrt(r)) on ln p: 0.420
- B: group ops / sqrt(r) per prime: 211: 110.1, 419: 157.7, 821: 347.4, 1619: 209.1, 3203: 177.1, 6421: 97.5, 12841: 154.1, 25601: 1558.4, 51203: 722.2
- B: slope of ln(group ops / sqrt(r)) on ln p: 0.292